

Week 6: Policy Gradient Methods

Seasons of Code — Reinforcement Learning

Policy Gradient Theorem

We directly parameterize a stochastic policy $\pi_\theta(a \mid s)$ and optimize the expected return

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \gamma^t r_t \right], \quad (1)$$

where $\tau = (s_0, a_0, s_1, a_1, \dots)$ is a trajectory sampled by running π_θ .

Derivation. Write the trajectory return as $R(\tau) = \sum_t \gamma^t r_t$ and the trajectory probability under π_θ as

$$p_\theta(\tau) = p(s_0) \prod_{t=0}^T \pi_\theta(a_t \mid s_t) p(s_{t+1} \mid s_t, a_t). \quad (2)$$

Then

$$J(\theta) = \int p_\theta(\tau) R(\tau) d\tau, \quad \nabla_\theta J(\theta) = \int \nabla_\theta p_\theta(\tau) R(\tau) d\tau. \quad (3)$$

Using the *log-derivative trick* $\nabla_\theta p_\theta(\tau) = p_\theta(\tau) \nabla_\theta \log p_\theta(\tau)$:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [\nabla_\theta \log p_\theta(\tau) R(\tau)]. \quad (4)$$

Since $\log p_\theta(\tau) = \log p(s_0) + \sum_t \log \pi_\theta(a_t \mid s_t) + \sum_t \log p(s_{t+1} \mid s_t, a_t)$, and the dynamics terms $p(s_0)$, $p(s_{t+1} \mid s_t, a_t)$ do not depend on θ , they vanish under ∇_θ . This gives the **Policy Gradient Theorem**:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\left(\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t \mid s_t) \right) R(\tau) \right] \quad (5)$$

This is the key result: we can compute a gradient of the expected return *without* knowing the environment dynamics $p(s_{t+1} \mid s_t, a_t)$, since they dropped out entirely.

Using causality (rewards before time t don't depend on a_t), this can be tightened to the per-timestep form:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t \mid s_t) G_t \right], \quad G_t = \sum_{k=t}^T \gamma^{k-t} r_k. \quad (6)$$

REINFORCE Algorithm

REINFORCE (Williams, 1992) is the Monte Carlo instantiation of the policy gradient theorem: run full episodes, compute the empirical return G_t at each timestep, and take a gradient ascent step.

Algorithm 1 REINFORCE

```
1: Initialize policy parameters  $\theta$ 
2: for each episode do
3:   Generate trajectory  $s_0, a_0, r_0, \dots, s_T, a_T, r_T$  using  $\pi_\theta$ 
4:   for  $t = 0, \dots, T$  do
5:      $G_t \leftarrow \sum_{k=t}^T \gamma^{k-t} r_k$ 
6:      $\theta \leftarrow \theta + \alpha \gamma^t G_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)$ 
7:   end for
8: end for
```

REINFORCE is an unbiased estimator of $\nabla_\theta J(\theta)$, but suffers from **high variance**: G_t is a single Monte Carlo sample of the return, and its scale (and even sign) can vary wildly across episodes, especially in sparse or noisy-reward environments, leading to slow and unstable learning.

Variance Reduction with a Baseline

We can subtract any state-dependent baseline $b(s_t)$ from G_t without introducing bias:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t \mid s_t) (G_t - b(s_t)) \right]. \quad (7)$$

Proof of unbiasedness. It suffices to show $\mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(a_t \mid s_t) b(s_t)] = 0$ for any b independent of a_t :

$$\mathbb{E}_{a_t \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t \mid s_t) b(s_t)] = b(s_t) \int \pi_\theta(a_t \mid s_t) \nabla_\theta \log \pi_\theta(a_t \mid s_t) da_t \quad (8)$$

$$= b(s_t) \int \nabla_\theta \pi_\theta(a_t \mid s_t) da_t \quad (9)$$

$$= b(s_t) \nabla_\theta \int \pi_\theta(a_t \mid s_t) da_t \quad (10)$$

$$= b(s_t) \nabla_\theta (1) = 0. \quad (11)$$

Since this holds in expectation for every t and every choice of $b(s_t)$, subtracting the baseline does not change $\mathbb{E}[\nabla_\theta J(\theta)]$ — the estimator remains unbiased.

Why it reduces variance. A common choice is $b(s_t) = V^{\pi_\theta}(s_t)$, the state-value function. Then $G_t - b(s_t)$ is an estimate of the **advantage** $A(s_t, a_t) = Q^{\pi_\theta}(s_t, a_t) - V^{\pi_\theta}(s_t)$: how much better action a_t is compared to the average action at s_t . This centers the learning signal around zero — actions better than average get a positive update, worse-than-average actions get a negative update, rather than every action being pushed in the same direction whenever G_t happens to be uniformly large or small (e.g. due to reward scale). Formally, for a random variable X and constant-in-expectation shift b ,

$$\text{Var}[X - b] = \text{Var}[X] - 2 \text{Cov}(X, b) + \text{Var}[b], \quad (12)$$

and choosing $b(s_t) \approx \mathbb{E}[G_t \mid s_t]$ makes b highly correlated with G_t , so the covariance term dominates and drives variance down, without touching the mean of the gradient estimator.

REINFORCE with Baseline

Algorithm 2 REINFORCE with Baseline

```
1: Initialize policy parameters  $\theta$ , value function parameters  $w$ 
2: for each episode do
3:   Generate trajectory  $s_0, a_0, r_0, \dots, s_T, a_T, r_T$  using  $\pi_\theta$ 
4:   for  $t = 0, \dots, T$  do
5:      $G_t \leftarrow \sum_{k=t}^T \gamma^{k-t} r_k$ 
6:      $\delta_t \leftarrow G_t - V_w(s_t)$  ▷ advantage estimate
7:      $w \leftarrow w + \beta \delta_t \nabla_w V_w(s_t)$  ▷ fit baseline via regression
8:      $\theta \leftarrow \theta + \alpha \gamma^t \delta_t \nabla_\theta \log \pi_\theta(a_t | s_t)$ 
9:   end for
10: end for
```
